Data Quality & Reconciliation

Period to 2026-09-30 · Demo Client

Prepared for	Demo Client
Engagement	DEMO-0010
Report date	11 November 2026
Version	1.0
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Run ID	dq_20261111T180124Z_aebd97

Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. The company, the customers, the banks and the transactions are fictitious.

ProQuant Analytics reviewed the finance data of Demo Client to answer two questions before anyone relies on the numbers: is the data fit for use, and do the systems agree? Quality was measured on the files exactly as received, with 23 rules and control totals that prove whether the extracts are complete. The ledger was then matched with the bank at 2026-09-30, every open item was diagnosed, and each account was proved to the cent.

Overall view: Data and reconciliations cannot yet be relied on

3 of 22 tests passed, 6 with a warning, 13 failed. Conclusions are drafted from the figures and reviewed by the analyst before release.

DATA-QUALITY SCORE	RECORDS MATCHED	ACCOUNTS EXPLAINED
96.5	98.1%	3 of 4
23 rules · 7 fail	13,550 of 13,810	13k USD unexplained

OPEN ITEMS, NOT TIMING	OLDER THAN 90 DAYS	COMPLETE EXTRACTS
151	58	2 short
407k USD · 78 timing	89% older than 30 days	1 day(s) fail the balance proof

Conclusions

- **Headline** 7,378 ledger entries and 6,940 bank lines on 4 accounts were reconciled at 2026-09-30: 98.1% of the records and 97.1% of the value matched with no break; 229 items stand between the books and the bank, worth USD 638k, of which USD 407k are not timing.
- **Completeness** The extracts are not complete: ledger 2026-08 is 23 rows short of its control total; bank 2026-07 is 5 rows short of its control total. The bank balance fails to roll forward on 1 day(s).
- **Data quality** The data-quality score is 96.5 of 100 on 23 rules; 7 rule(s) fail, 1 of them critical. Worst rules: Reference is in the agreed form (3,795 rows); Issue date is in ISO form (1,839 rows); Invoice points to an active customer (354 rows).
- **Master data** 52 pairs of customers look like duplicates (6.2% of the master) and 25 IBANs fail their check digits. A bad IBAN is a payment that fails or reaches the wrong account.
- **Migration** Since the move to ERP-B 35.2% of the rows break a critical or major rule, against 0.7% on ERP-A; the match rate of ledger entries fell from 98.1% to 97.9%. Dates, reference formats and blank fields explain it.
- **Statements** 3 of 4 accounts are fully explained. BA-EUR-01 has +11,629.54 EUR unexplained.
- **Breaks** Largest causes of real breaks: Wrong account (12 items, USD 106k); Duplicate posting (14 items, USD 76k); Transposed digits (8 items, USD 62k). 58 item(s) are older than 90 days. 78 items are only timing and clear on their own.
- **Receivables** The subledger and the control account differ by USD 14k at the period end; the difference first appears in 2026-08.

- **Effort** Clearing these breaks takes about 9 hours a month, about USD 4,711 a year. The three largest causes are 44% of the hours.
- **Overall** Data and reconciliations cannot yet be relied on: 3 tests passed, 6 with a warning, 13 failed. Priority actions: Ask for the missing rows and add a control total to every extract. Ask the bank for the missing statement lines; the balance proves they exist.
- **Analyst** Illustrative note (synthetic data): the finance controller confirms that the interface was re-run by hand on 14 August after a time-out.
- **Analyst** Illustrative note (synthetic data): the bank has been asked for the BA-EUR-01 statement of 22 July.

Actions

Test	Status	Action
Months where an extract differs from its control total	FAIL	Ask for the missing rows and add a control total to every extract.
Days where the bank balance does not roll forward	FAIL	Ask the bank for the missing statement lines; the balance proves they exist.
Customers with a likely duplicate	FAIL	Review the candidate pairs with the master-data owner and merge.
Customers with an invalid IBAN	FAIL	Correct the IBANs and validate them at entry; a bad IBAN is a failed or misdirected payment.
Failure rate of the new system against the old	FAIL	Fix the ERP-B interface: date format, reference format, mandatory fields.
Accounts with an unexplained difference	FAIL	Do not sign this account until the difference is explained.
Receivables subledger against the control account	FAIL	Find the manual journal behind the difference and post it through the subledger.
Open errors and unknowns against the value of the period	FAIL	Clear the break tracker by priority, with owners and dates.
Open items older than 30 days	FAIL	Review aged items weekly until they are gone.
Open items older than 90 days (not timing)	FAIL	Escalate items over 90 days to the finance controller.
Unidentified bank items	FAIL	Identify the senders; hold in suspense and follow the unclaimed-funds policy.
Duplicate postings	FAIL	Reverse the second postings and add a duplicate check to the interface.
Items at the bank never booked	FAIL	Book the items and add a monthly routine for charges and interest.
Extracts delivered late	WARN	Agree delivery dates with the source owners and add an alert for a late file.
Overall data-quality score	WARN	Start with the rules that fail most, by owner; see the issue list.
Critical rules that fail	WARN	Fix the critical rules first: they break the reconciliation or the payment.
Invoices with an unknown customer	WARN	Create or correct the customers and block invoices without a valid customer.
Value matched with no break	WARN	Clear the largest open items first.
Accounts fully explained	WARN	Explain the open difference and sign each statement.

Source: tables/tests.csv

Methodology

This report tells what was found. How each number is built, with formulas and worked examples, is in the methodology document. Two things help when reading on: quality was measured before anything was cleaned, so a problem we repaired still counts; and every account is shown as books, bank and what lies between, down to the last cent.

Results

1. Did everything arrive?

Source	Month	Expected rows	Received	Rows	Expected total	Received total	Total
ledger	2026-08	1,195	1,172	-23	2,565,815.96	2,639,911.16	74,095.20
bank	2026-07	1,308	1,303	-5	2,448,494.01	2,436,864.47	-11,629.54

Source: tables/control_totals_check.csv

The extracts that differ from their control totals. In every other month the rows and the total agree.

Balance proof

Account	Day	Opening	Lines in the file	Closing per the bank	Missing
BA-EUR-01	2026-07-22	2,059,695.36	0.00	2,071,324.90	11,629.54

Source: tables/balance_proof_failures.csv

On these days the bank's balance moves by more than the lines in the file: lines exist that we did not receive.

2. Is the data fit for use?

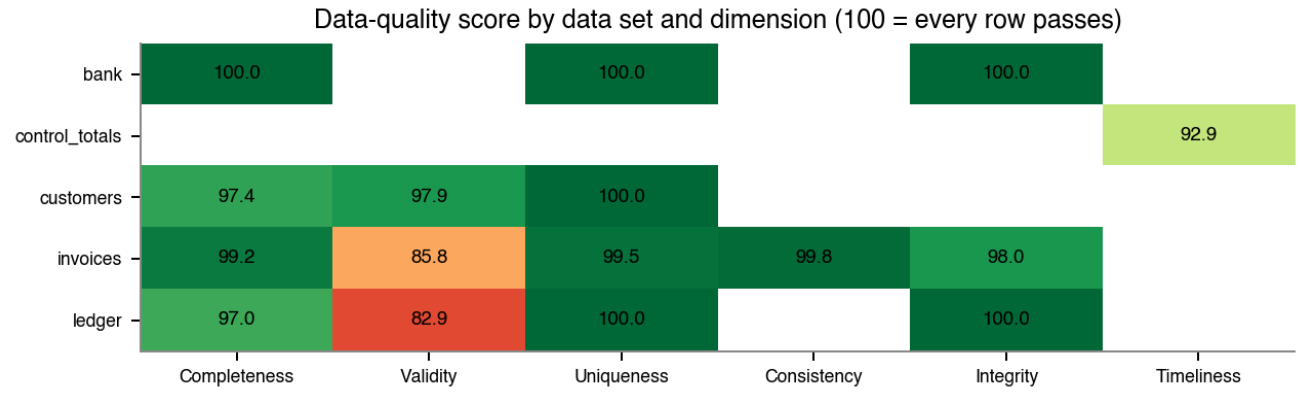


Figure 1. Data-quality score. Source: tables/dq_scorecard.csv

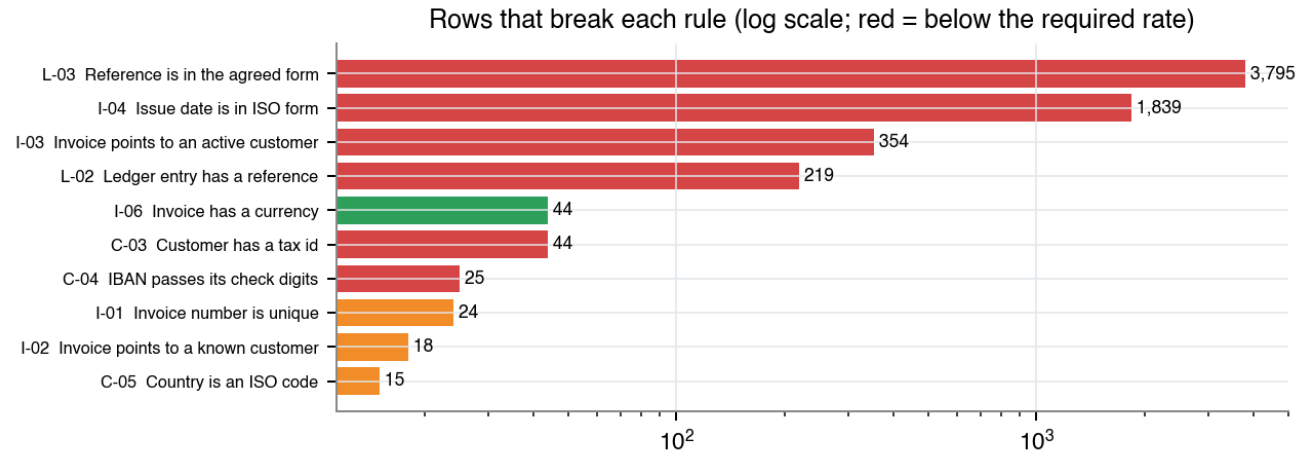


Figure 2. Rules with the most failures. Source: tables/dq_results.csv

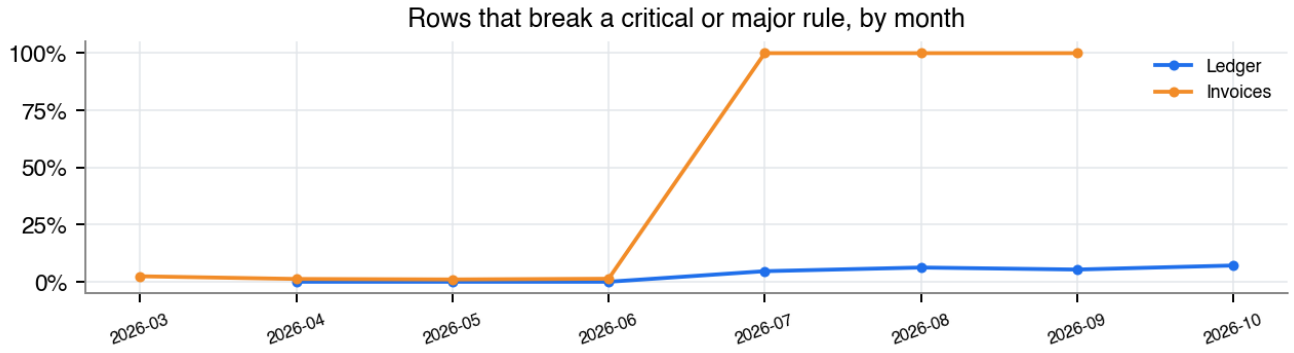


Figure 3. Quality by month: the change comes with the move to the new system. Source: tables/dq_trend.csv

Rule	Data set	Dimension	What it checks	Severity	Failed	Passed	Required	Status
L-03	ledger	Validity	Reference is in the agreed form	minor	3,795	48.56%	95.0%	FAIL
I-04	invoices	Validity	Issue date is in ISO form	major	1,839	64.72%	99.0%	FAIL
I-03	invoices	Integrity	Invoice points to an active customer	minor	354	93.21%	95.0%	FAIL
L-02	ledger	Completeness	Ledger entry has a reference	major	219	97.03%	99.0%	FAIL
C-03	customers	Completeness	Customer has a tax id	major	44	94.76%	99.0%	FAIL
C-04	customers	Validity	IBAN passes its check digits	critical	25	97.02%	99.9%	FAIL
T-01	control_totals	Timeliness	Extract delivered by the agreed date	major	1	92.86%	99.0%	FAIL
I-06	invoices	Completeness	Invoice has a currency	major	44	99.16%	99.0%	PASS
I-07	invoices	Consistency	Due date is not before the issue date	major	10	99.81%	99.0%	PASS
I-08	invoices	Consistency	A paid invoice has a payment date	major	9	99.83%	99.0%	PASS
C-01	customers	Uniqueness	Customer id is unique	critical	0	100.00%	99.9%	PASS
C-02	customers	Completeness	Customer has a name	major	0	100.00%	99.0%	PASS
C-06	customers	Validity	Status is a known value	minor	0	100.00%	95.0%	PASS
L-01	ledger	Uniqueness	Ledger entry id is unique	critical	0	100.00%	99.9%	PASS
L-04	ledger	Integrity	Ledger entry is on a known bank account	critical	0	100.00%	99.9%	PASS
L-05	ledger	Validity	Value date is in ISO form	major	0	100.00%	99.0%	PASS
B-01	bank	Uniqueness	Bank line id is unique	critical	0	100.00%	99.9%	PASS
B-02	bank	Completeness	Bank line has a narrative	minor	0	100.00%	95.0%	PASS
B-03	bank	Integrity	Bank line is on a known account	critical	0	100.00%	99.9%	PASS
I-01	invoices	Uniqueness	Invoice number is unique	critical	24	99.54%	99.9%	WARN
I-02	invoices	Integrity	Invoice points to a known customer	critical	18	99.65%	99.9%	WARN
C-05	customers	Validity	Country is an ISO code	major	15	98.21%	99.0%	WARN
I-05	invoices	Validity	Amount is positive and below the plausible maximum	critical	9	99.83%	99.9%	WARN

Source: tables/dq_results.csv

3. Customer master

52 candidate pairs of duplicate customers. These are candidates: merging is the decision of the data owner.

Customer	Name	Twin	Name of twin	Similarity	Why
C-0522	Pioneer Industries 50 LLC	C-9005	Pioneer Industries 50 L.L.C.	100%	same IBAN
C-0169	Vertex Marine 85 Ltd	C-9002	Vertex Marine 85 Ltd.	100%	same IBAN
C-0145	Kestrel Interiors 50 GmbH	C-9003	Kestrel Interiors 50 GmbH	100%	same IBAN
C-0299	Amber Textiles 78 LLC	C-9037	Amber Textiles 78 LLC	100%	same IBAN
C-0515	Granite Motors 47 Ltd	C-9001	Granite Motors 47 Ltd	100%	same IBAN
C-0406	Crescent Packaging 32 LLC	C-9025	Crescent Packaging 32 LLC.	100%	same IBAN
C-0652	Summit Interiors 48 GmbH	C-9040	Summit Interiors 48 GmbH.	100%	same IBAN
C-0124	Northern Motors 38 Ltd	C-9004	NORTHERN MOTORS 38 LTD	100%	same IBAN

Source: tables/duplicate_customers.csv

4. Do the systems agree?

Account	Ccy	Books	Bank	Difference	Explained	Unexplained	Items	Status
BA-USD-01	USD	4,747,488.69	4,842,737.23	95,248.54	95,248.54	0.00	77	Reconciled, open items
BA-USD-02	USD	-2,660,309.73	-2,591,847.89	68,461.84	68,461.84	0.00	50	Reconciled, open items
BA-EUR-01	EUR	3,667,350.48	3,600,639.89	-66,710.59	-78,340.13	11,629.54	55	Not reconciled
BA-AED-01	AED	8,030,631.78	7,969,323.61	-61,308.17	-61,308.17	0.00	47	Reconciled, open items

Source: tables/statements.csv

The statement of each account, line by line, with sign-off blocks, is delivered as a separate document.

Pass	Pairs	Ledger entries	Value	Mean days apart
Reference and amount	5,536	5,536	21,067,312	1.0
Amount and date	1,069	1,069	3,810,077	1.1
Batch	213	661	1,942,180	1.0
Within tolerance	77	77	328,141	0.9

Source: tables/match_by_pass.csv

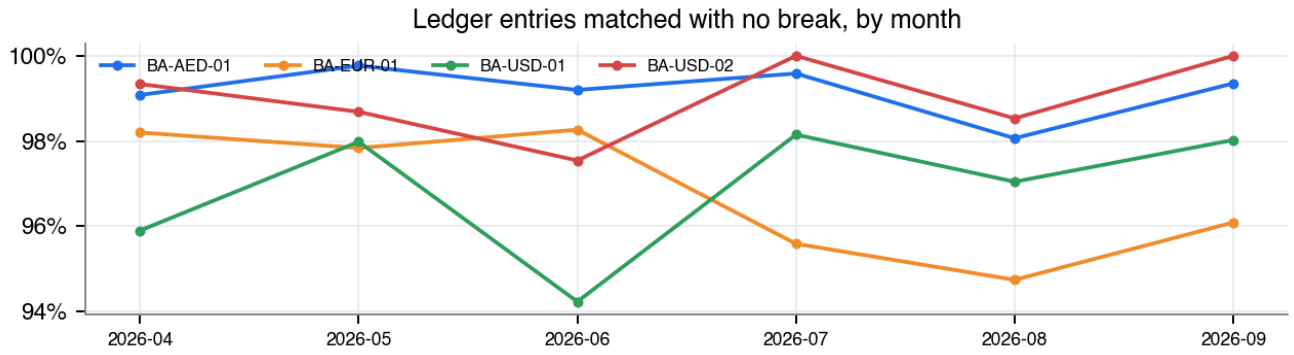


Figure 4. Match rate by month and account. Source: tables/match_by_month.csv

5. What stands between the books and the bank

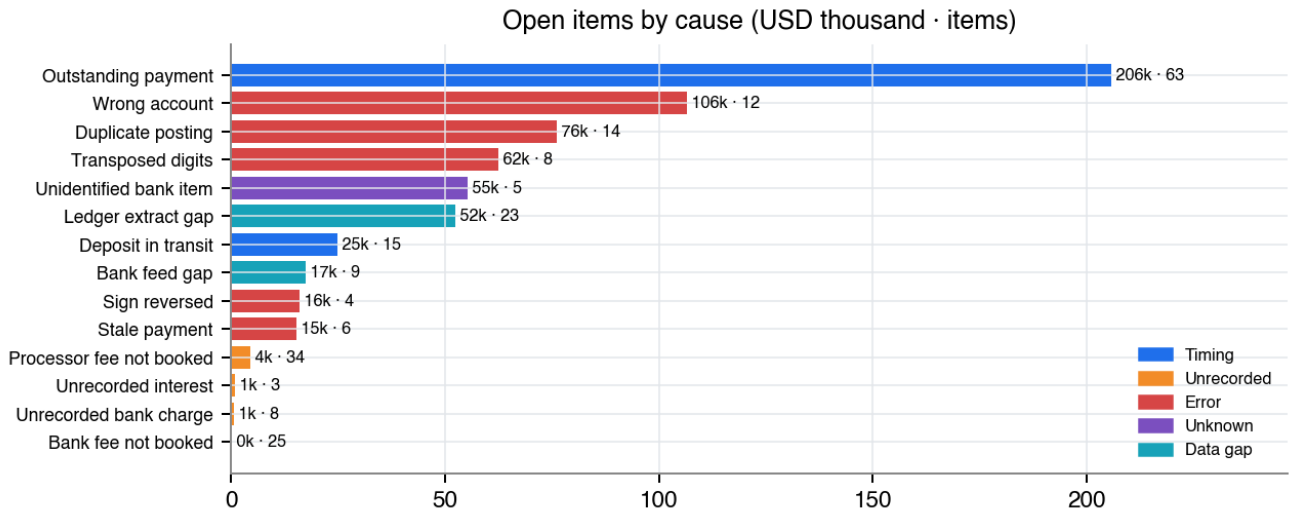


Figure 5. Open items by cause. Source: tables/break_causes.csv

Cause	Nature	Items	USD	Oldest (days)	Hours
Outstanding payment	Timing	63	205,869	6	2.1
Wrong account	Error	12	106,473	163	5.0
Duplicate posting	Error	14	76,100	47	5.8
Transposed digits	Error	8	62,388	142	3.3
Unidentified bank item	Unknown	5	55,371	134	3.8
Ledger extract gap	Data gap	23	52,311	57	11.5
Deposit in transit	Timing	15	24,902	1	0.5
Bank feed gap	Data gap	9	17,378	72	4.5
Sign reversed	Error	4	15,950	155	1.7
Stale payment	Error	6	15,316	174	2.5
Processor fee not booked	Unrecorded	34	4,421	175	5.7
Unrecorded interest	Unrecorded	3	934	122	0.5
Unrecorded bank charge	Unrecorded	8	549	153	1.3
Bank fee not booked	Unrecorded	25	259	169	4.2

Source: tables/break_causes.csv

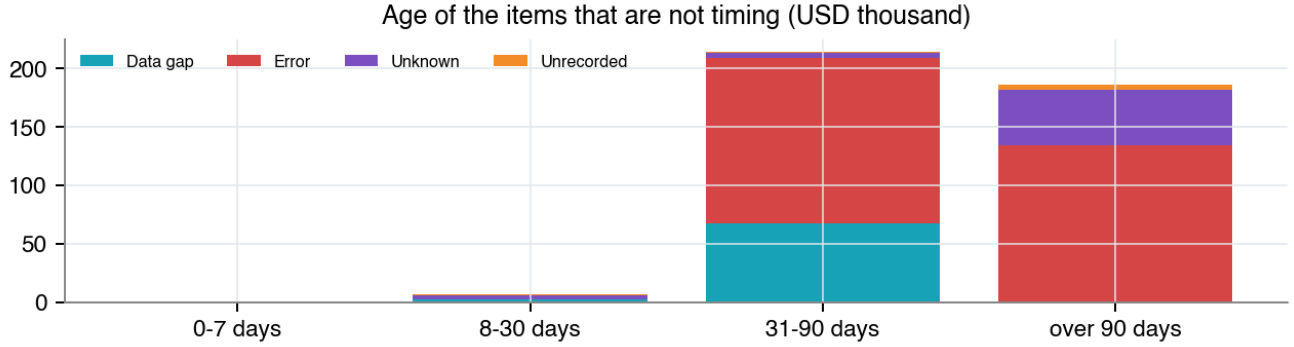


Figure 6. Age of the items that are not timing. Source: tables/break_ageing.csv

Owner	Priority	Items	USD
Data owner	High	32	69,689
Finance	High	65	250,779
Finance	Low	38	3,188
Finance	Medium	11	28,424
Treasury	High	1	48,250
Treasury	Medium	4	7,121

Source: tables/break_owners.csv

The fifteen first items

Item	Account	Cause	Date	Age	Amount	Reference	Priority	Owner
BR-00001	BA-EUR-01	Transposed digits	2026-05-12	141	-54,000.00	INV-01618	High	Finance
BR-00002	BA-USD-01	Unidentified bank item	2026-05-19	134	48,250.00		High	Treasury
BR-00003	BA-USD-02	Wrong account	2026-08-10	51	22,998.87		High	Finance
BR-00004	BA-USD-01	Wrong account	2026-08-10	51	22,998.87		High	Finance
BR-00005	BA-EUR-01	Duplicate posting	2026-08-14	47	18,195.60	INV-03901	High	Finance
BR-00006	BA-EUR-01	Duplicate posting	2026-08-14	47	12,367.37	INV-03922	High	Finance
BR-00007	BA-USD-02	Stale payment	2026-05-08	145	-13,066.45	PAY-01318	High	Finance
BR-00008	BA-AED-01	Duplicate posting	2026-08-14	47	38,881.25		High	Finance
BR-00009	BA-EUR-01	Duplicate posting	2026-08-14	47	9,457.81		High	Finance
BR-00010	BA-USD-01	Ledger extract gap	2026-08-26	35	-9,042.43	PAY-01627	High	Data owner
BR-00011	BA-USD-02	Wrong account	2026-05-25	128	8,661.30	INV-04828	High	Finance
BR-00012	BA-USD-01	Wrong account	2026-05-27	126	8,661.30	INV-04828	High	Finance
BR-00013	BA-USD-02	Ledger extract gap	2026-08-08	53	-7,938.19	PAY-01620	High	Data owner
BR-00014	BA-USD-02	Wrong account	2026-06-01	121	7,728.28	INV-01588	High	Finance
BR-00015	BA-USD-01	Wrong account	2026-06-03	119	7,728.28	INV-01588	High	Finance

Source: tables/breaks.csv

The full list is in the break tracker, with owners and a status for each.

6. Receivables against the ledger

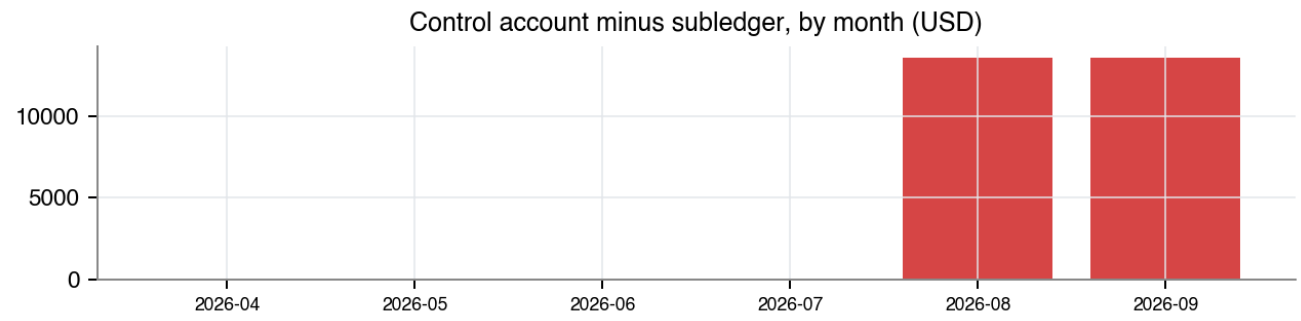


Figure 7. Control account against the invoices. Source: tables/tie_out.csv

Month	Subledger	Control account	Difference	Change	Status
2026-04	2,569,464.20	2,569,464.20	0.00	0.00	PASS
2026-05	3,089,314.17	3,089,314.17	0.00	0.00	PASS
2026-06	3,282,737.68	3,282,737.68	-0.00	-0.00	PASS
2026-07	3,271,343.04	3,271,343.04	-0.00	0.00	PASS
2026-08	3,447,353.26	3,460,984.81	13,631.55	13,631.55	FAIL
2026-09	1,195,989.30	1,209,620.84	13,631.54	-0.01	FAIL

Source: tables/tie_out.csv

A difference that appears in one month and then stays is typically a manual journal to the control account that never reached the subledger. Invoices above the plausible maximum are left out of the subledger here and listed under rule I-05.

7. What it costs

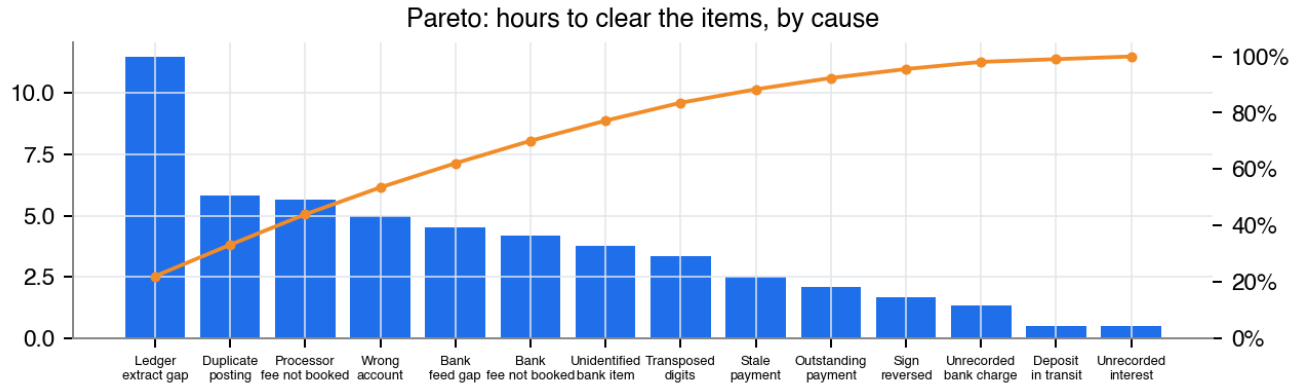


Figure 8. Hours to clear the items, by cause. Source: tables/break_causes.csv

About 9 hours a month for these items. This counts only the clearing of the items, on the minutes agreed at kickoff; it does not count the cost of a wrong number reaching a decision.

8. Rows that could not be used

No row had to be excluded.

9. All tests

Area	Test	Result	Threshold	Status
Completeness	Months where an extract differs from its control total	2 month(s)	warn above 0 month(s), fail above 0 month(s)	FAIL
Completeness	Days where the bank balance does not roll forward	1 day(s)	warn above 0 day(s), fail above 0 day(s)	FAIL
Completeness	Extracts delivered late	7%	warn above 0%, fail above 25%	WARN
Completeness	Rows that could not be used	0.00%	warn above 0.00%, fail above 1.00%	PASS
Data quality	Overall data-quality score	96.5	warn below 99.0, fail below 95.0	WARN
Data quality	Critical rules that fail	1 rule(s)	warn above 0 rule(s), fail above 2 rule(s)	WARN
Data quality	Customers with a likely duplicate	6.2%	warn above 1.0%, fail above 5.0%	FAIL
Data quality	Customers with an invalid IBAN	3.0%	warn above 0.0%, fail above 2.0%	FAIL
Data quality	Invoices with an unknown customer	0.35%	warn above 0.00%, fail above 1.00%	WARN
Data quality	Failure rate of the new system against the old	47.4x	warn above 1.5x, fail above 3.0x	FAIL
Reconciliation	Records matched with no break	98.1%	warn below 97.0%, fail below 90.0%	PASS
Reconciliation	Value matched with no break	97.1%	warn below 98.0%, fail below 92.0%	WARN
Reconciliation	Accounts with an unexplained difference	1 account(s)	warn above 0 account(s), fail above 0 account(s)	FAIL
Reconciliation	Accounts fully explained	75%	warn below 100%, fail below 75%	WARN
Reconciliation	Receivables subledger against the control account	13,632 USD	warn above 0 USD, fail above 10,000 USD	FAIL
Breaks	Open errors and unknowns against the value of the period	2.08%	warn above 0.50%, fail above 2.00%	FAIL
Breaks	Open items older than 30 days	88.7%	warn above 10.0%, fail above 30.0%	FAIL
Breaks	Open items older than 90 days (not timing)	58 item(s)	warn above 0 item(s), fail above 5 item(s)	FAIL
Breaks	Unidentified bank items	55,371 USD	warn above 0 USD, fail above 25,000 USD	FAIL
Breaks	Duplicate postings	76,100 USD	warn above 0 USD, fail above 20,000 USD	FAIL
Breaks	Items at the bank never booked	70 item(s)	warn above 3 item(s), fail above 30 item(s)	FAIL
Effort	Manual hours a month to clear the breaks	9 h	warn above 20 h, fail above 60 h	PASS

Source: tables/tests.csv

Limitations and assumptions

- Control totals are as good as their source: they prove the file agrees with the system, not that the system is right.
- A match is evidence that two records describe one payment, not proof. The loosest passes can pair two unrelated payments of the same amount; each pair records its pass so that it can be reviewed.
- The diagnosis of a break is a best reading of the data. The owner confirms it against the source document.
- Duplicate customers are candidates for review. Two similar names can be two companies.
- Rules measure what they were written to measure. A value can pass every rule and still be wrong.
- Bank data ends a few days after the period end; an item that clears later is shown as open.

Appendix

Cleaning log

step	action	rows_affected
harmonise	references put in one form (INV01234 -> INV-01234)	3688
harmonise	references read from the bank narrative	5828
duplicates	invoice rows sent twice removed	12
harmonise	invoice dates read from dd/mm/yyyy as well as ISO	5261
harmonise	blank invoice currency taken from the customer master	44

Source: tables/cleaning_log.csv

Data quality

accounts · 4 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

bank · 6940 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

bank_balances · 772 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

control_totals · 14 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

customers · 840 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	tax_id	44	44 missing (5.24%)

Source: data_quality/dq_customers.json

fx_rates · 27 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

gl_balances · 6 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

invoices · 5212 rows · PASSED (0 errors, 1 warnings)

Check	Severity	Column	Count	Message
missing_values	info	currency	44	44 missing (0.84%)
missing_values	info	paid_date	257	257 missing (4.93%)
duplicate_rows	warning		12	12 fully duplicated rows

Source: data_quality/dq_invoices.json

ledger · 7378 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	reference	219	219 missing (2.97%)
missing_values	info	batch_ref	6712	6712 missing (90.97%)

Source: data_quality/dq_ledger.json

ledger_clean · 7378 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	reference	219	219 missing (2.97%)
missing_values	info	batch_ref	6712	6712 missing (90.97%)
missing_values	info	ref	219	219 missing (2.97%)

Source: data_quality/dq_ledger_clean.json

Audit trail

Run ID	`dq_20261111T180124Z_aebd97`
Run started / finished (UTC)	2026-11-11T18:01:24+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`2703b230a84f (uncommitted changes) · core 0.1.0`
Config SHA-256	`1ede7114dffc56f9`
Input: customers	`840 rows · sha256 1cda64a6c8808d34`
Input: invoices	`5212 rows · sha256 7fdf1307089369a6`
Input: ledger	`7378 rows · sha256 520f9294fe16a6d2`
Input: bank	`6940 rows · sha256 405588f5b1cb031d`
Input: bank_balances	`772 rows · sha256 b98ce8755b81d1fe`
Input: accounts	`4 rows · sha256 f01ea25ec9666d1c`
Input: control_totals	`14 rows · sha256 cc9e55ee18dc01c4`
Input: gl_balances	`6 rows · sha256 35f3ddd2b64253ad`
Input: fx_rates	`27 rows · sha256 cba0676dbf3ce091`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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